

SOFTWARE TESTING

MASTERCLASS

Test Planning and Estimation



Advance Testing Concepts

Test Planning and Estimation

Test planning is a vital process that involves creating a document that outlines the goals, resources, and steps for conducting a specific test on a software or hardware product



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Test Planning and Estimation

The purpose of test planning is to ensure that the testing effort is well-organised and effective

Creating a test plan can take several days or even weeks

It's important to define a test plan before starting testing

What
should
we test ?

How
should
we test ?

How much
time should
be allocated
for testing ?





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Test Planning and Estimation

Test Management includes:

- Test Organization
- Test Planning and Estimation
- Test Monitoring and Control
- Configuration Management
- Defect Management





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Test Planning Steps
Determining the scope and identifying risks, along with establishing the testing objectives
Defining the overall approach for testing, including the test levels and the criteria for entry and exit
Integrating and coordinating the testing activities with the software life cycle activities
Making decisions regarding what should be tested, which roles will perform the testing activities, how the activities should be carried out, and how the test results will be evaluated
Scheduling the test analysis and design activities



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Test Planning Steps
Establishing schedules for test implementation, execution, and evaluation
Assigning resources for the different defined activities
Defining the necessary amount of detail, structure, and templates for the test documentation
Selecting appropriate metrics for monitoring and controlling the preparation and execution of tests, defect resolution, and managing risk
Determining the level of detail required for test procedures to ensure reproducibility in test preparation and execution

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Entry criteria define the conditions or requirements that must be met to initiate testing, such as starting a new test level or beginning the execution of a set of tests

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Entry criteria's:

- Availability and readiness of the test environment
- Readiness of test tools in the test environment
- Availability of testable code
- Availability of test data

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Exit criteria define the conditions or requirements that indicate when testing should be completed, such as reaching the end of a test level or achieving specific objectives for a set of tests



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Exit criteria's:

- Estimates of defect density or reliability measures
- Cost considerations
- Remaining risks, such as unresolved defects or insufficient test coverage in certain areas
- Measures of completeness, such as code coverage, functionality coverage, or risk coverage



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Estimation involves making predictions or rough estimations of the cost or duration of a task. In the context of testing, time estimation is a crucial aspect of project planning

Setting
realistic
targets

Allocating
budgets
and
resources

Determining
project
timelines



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Scrum
Poker

One effective approach for estimation is to consider the complexity of the task